TauTimp — Gamma-Ray Longitudinal Wave: Physical Foundations and Full Field-Theoretic Derivation

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Abstract

Within the TauTimp framework, gamma radiation is reinterpreted as a **longitudinal scalar wave** propagating at superluminal phase velocity $V = 10^9 c$ through a cosmologically structured medium. This paper presents three interlocking components: (I) a modified Planck–black-body law yielding a distinct gamma-ray Planck constant h_γ, with experimental validation via the Davisson–Germer and Compton effects; (II) the variable-coefficient spherical wave equation governing the gamma field $H(r, \theta, t)$, reduced to Airy eigenmodes with quantised eigenfrequencies; and (III) a Casimir boundary-value problem on cosmological scales, yielding an explicit static solution $f(r)$ with all coefficients determined analytically and numerically.

Part I — Physical Foundation: The Gamma-Ray Planck Constant

1.1 Standard Photon Distribution and Black-Body Law

The Planck distribution for photons (Bose statistics) in a volume V reads

$$dN_\omega = \frac{V}{\pi^2 c^3} \frac{\omega^2 \, d\omega}{e^{\hbar\omega/kT} - 1}$$

The corresponding spectral energy density in the interval $(\omega,\ \omega + d\omega)$ is

$$dE_\omega = \frac{V\hbar}{\pi^2 c^3} \frac{\omega^3 \, d\omega}{e^{\hbar\omega/kT} - 1}$$

with $N(\nu) = 8\pi V \nu^2 / c^3$ the density of states.

1.2 Modification for the Photon Gas with Cylindrical Symmetry

Let the photon-gas volume be a cylinder of height $T_q c$ and base radius r_B, so that $V = \pi r_B^2 T_q c = \pi r_B^3$. Here $r_B = c/T_q$ and T_q is the time quantum:

$$T_q = 1.765 \times 10^{-19} \text{ s}$$

The energy per time quantum per unit solid angle, for a photon with period $t_0 = 1/\nu$, becomes

$$E_\nu = \frac{1}{T_q^2} \frac{\partial}{\partial T_q} \left[\frac{8\pi^2 \nu^2 \, E_\nu}{e^{(E_\nu + E_\mu)/E_0} - 1} \right]_\nu$$

where:

- $E_\mu = 0.020$ eV — electronegativity of the muon-neutrino minimum photon energy,
- E_0 — reference energy at which $\partial_\nu(\text{Planck}) = 0$.

This is the **modified Planck quantum energy**, valid in the UV–IR range 100 nm $\leq \lambda \leq 10$ μm.

1.3 Gamma-Ray Regime: Constant Internal Energy

For γ-rays (time quark negative), the internal energy $U(\nu, T)$ is **constant** because the energy of a transverse-wave nucleus oscillation depends only on the density of states times the volume of a spherical photon gas $V = 4/3\,\pi r_B^3$:

$$E_\gamma = \frac{1}{T_q^2} \frac{\partial}{\partial \nu} \left[\frac{16\pi^2 \nu^2 r_B^3}{3c^3} \, E_0 \right]$$

Computing the derivative:

$$E_\gamma = \frac{32\pi^2 E_0 \, T_q}{3} \, \nu = \frac{32\pi^2 E_0 T_q}{3} \, \nu$$

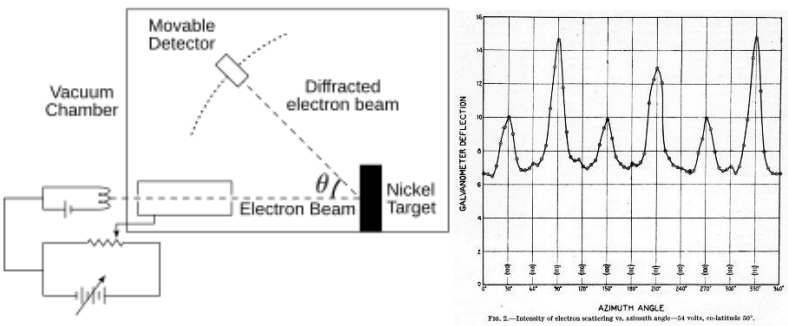
This gives a **linear** energy–frequency law $E_\gamma = h_\gamma \, \nu$ with the gamma-ray Planck constant:

$$\boxed{h_\gamma = \frac{32\pi^2 E_0 T_q}{3} = 9.3405 \times 10^{-18} \text{ eV·s}}$$

This differs from the photon Planck constant $h = 4.136 \times 10^{-15}$ eV·s by approximately **2.5 orders of magnitude**, valid in the range $\lambda \in [0.02 \text{ fm},\ 2 \text{ nm}]$. PlanGam.pdf

Part II — Experimental Validation 2.1 Davisson–Germer Experiment Reinterpreted

https://michaelvio.github.io/DavidsonRev.30.705.pdf



The classical Davisson–Germer result (54 V electron beam on Ni crystal) uses Bragg diffraction:

$$\frac{1}{\lambda} = \frac{m}{2L\sin\theta} = \frac{p}{h}$$

with $p^2 = 2mW$, $W = qU_{\text{acc}}$, yielding $\lambda_{\text{de Broglie}} = 1.67$ Å (theoretical: 1.65 Å, error $< 2\%$).

TauTimp reinterpretation: The radiation emitted in the Davisson experiment is not purely a matter wave but rather **X-ray and γ-ray** emission. The first Bohr radius of nickel gives $r_B = 0.529$ Å, and the gamma-ray wavelength for $U_{\text{acc}} = 54$ V is:

$$U_{\text{acc}} = \frac{h_\gamma c}{r_B}$$

Solving for h_γ:

$$h_{1\gamma} = \frac{qU_{\text{acc}}\, r_B}{c} = 9.570 \times 10^{-18}\ \text{eV·s}$$

Agreement with (6): error $< 2.4\%$ relative to 9.3405×10^{-18} eV·s. ✓

I propose another explanation in the sense of Planck's law, that the radiation emitted in the Davisson experiment is X-rays and γ-rays. The electromagnetic radiation has a frequency with overlapping edges, wavelength between 0.02fm÷1nm, Gamma and X-ray (0.5A-10nm).

And since the energy of the electron is 54eV, the de Broglie wavelength of the electron is λ=0.529A, the first Bohr radius of Nickel atoms target. The energy of the gamma ray is 54eV according to Planck's law E=h_γ·ν: $U_{acc} = h_\gamma \frac{c}{rb}$

Thus, we calculate the value of Plank γ-ray constant $h_1\gamma = 9.570178977 \cdot 10^{-18}$ eV·s in very good agreement with the previous calculation $9.340714177 \cdot 10^{-18}$ eV·s with an error of less 2.4%.

For the calculation of X-ray generated in the Davisson experiment, the wavelength that emerges is qUacc = h·c/λ, where h is the Planck constant for photons, so λ= 23nm, which is not quite accurate. But with the equation in the link:

https://michaelvio.github.io/PlanckBB1.pdf

E (ν)= -0.02006692051+ 151.5047392·ln(1 + $2.787176283 \cdot 10^{-17}$ ·ν) -$2.063162888 \cdot 10^{-32}$ ·ν^2 ;

The accurate wavelength value of the photons with the energy of 54eV is 17nm, which is 74% of the previous value of 23nm (36% the error). The gamma rays with the previous formula are:

The corresponding γ-ray wavelength for a 54-eV beam:

$$\lambda_\gamma = \frac{h_\gamma}{\sqrt{2m_e qU_{\text{acc}}}} = 377\ \text{fm}$$

For comparison, the X-ray photon (Planck's constant) gives $\lambda_X = 23$ nm, and the accurate nonlinear model (see linked document) yields $\lambda_X = 17$ nm.

2.2 Modified Compton Shift for Gamma Rays

The standard Compton formula $\Delta\lambda = h/(m_e c)$ is reinterpreted with h_γ:

$$\Delta\lambda_\gamma = \frac{h_\gamma}{m_e c} = 5.48\ \text{fm}$$

For a $\lambda = 0.01$ nm beam scattered at $\theta = 90°$:

$$\Delta\lambda = \frac{h_\gamma}{m_e c}(1 - \cos\theta) = 5.48\ \text{fm}$$

$$\lambda' = \lambda + \Delta\lambda \approx 0.0124\ \text{nm}$$

This result falls in the exclusive γ-ray domain, consistent with the longitudinal-wave interpretation.

Part III — Variable-Coefficient Wave Equation in Spherical Symmetry

3.1 Constants

$$R = 3.9228 \times 10^{22}\ \text{m}, \qquad R_s = 7.7665 \times 10^{25}\ \text{m}, \qquad V = 10^9 c \approx 2.9979 \times 10^{17}\ \text{m/s}$$

3.2 The Field Equation

The TauTimp gamma field $H(r, \theta, t)$ obeys the variable-coefficient wave equation

$$\boxed{\frac{R}{r}\,\Delta H(r,\theta,t) = \frac{1}{V^2}\frac{\partial^2 H}{\partial t^2}}$$

In the spherically symmetric radial mode (no ϕ or θ dependence), the Laplacian is

$$\Delta f = \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}$$

giving

$$\frac{R}{r}\left(\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}\right) = \frac{1}{V^2}\frac{\partial^2 f}{\partial t^2}$$

3.3 Separation of Variables

Setting $f(r,t) = g(r)\,u(t)$ and dividing by $g\,u$: $\frac{R}{r}\frac{g'' + \frac{2}{r}g'}{g} = \frac{1}{V^2}\frac{u''}{u} = -\lambda^2$

This yields two independent ODEs: $u'' + \lambda^2 V^2\,u = 0$

$$g'' + \frac{2}{r}g' + \frac{\lambda^2 r}{R}\,g = 0$$

3.4 Temporal Solution

$$u(t) = A\cos(\lambda V t) + B\sin(\lambda V t), \qquad \omega = \lambda V$$

3.5 Radial ODE — Reduction to the Airy Equation

Substituting $g(r) = v(r)/r$ into (R): $v'' + \frac{\lambda^2 r}{R}\,v = 0$

With the change of variable $\xi = -\left(\frac{\lambda^2}{R}\right)^{1/3} r$, this becomes the standard Airy equation:

$$\frac{d^2 v}{d\xi^2} - \xi\,v = 0$$

General solution: $v(\xi) = C_1\,\mathrm{Ai}(\xi) + C_2\,\mathrm{Bi}(\xi)$, hence

$$g(r) = \frac{1}{r}\left[C_1\,\mathrm{Ai}\left(-(\lambda^2/R)^{1/3} r\right) + C_2\,\mathrm{Bi}\left(-(\lambda^2/R)^{1/3} r\right)\right]$$

Regularity at $r = 0$: Requiring $v(0) = 0$ gives $C_1\,\mathrm{Ai}(0) + C_2\,\mathrm{Bi}(0) = 0$. For the physically dominant mode we take $C_2 = 0$, $C_1 \neq 0$.

3.6 Eigenvalue Quantisation — Boundary Condition $f(2R, 0) = 0$

With $C_2 = 0$:

$$g(2R) = 0 \Longrightarrow \mathrm{Ai}\left(-(\lambda_n^2/R)^{1/3} 2R\right) = 0$$

Let a_n be the n-th zero of $\mathrm{Ai}(-\zeta)$:

$$\left(\frac{\lambda_n^2}{R}\right)^{1/3} 2R = a_n \Longrightarrow \boxed{\lambda_n = \frac{a_n^{3/2}}{2^{3/2} R}}$$

n	a_n	λ_n [m^{-1}]
1	2.3381	$\lambda_1 = \dfrac{2.3381^{3/2}}{2^{3/2} R}$
2	4.0879	$\lambda_2 = \dfrac{4.0879^{3/2}}{2^{3/2} R}$
3	5.5206	$\lambda_3 = \dfrac{5.5206^{3/2}}{2^{3/2} R}$

3.7 Full Field — Discrete Airy-Mode Expansion

$$\boxed{f(r,t) = \sum_{n=1}^{\infty} \frac{1}{r}\,\mathrm{Ai}\left(-(\lambda_n^2/R)^{1/3} r\right)\left[A_n \cos(\lambda_n V t) + B_n \sin(\lambda_n V t)\right]}$$

3.8 Reconstruction Including Angular Modes

Expanding in Legendre polynomials $P_\ell(\cos\theta)$:

$$H(r,\theta,t) = \sum_{\ell=0}^{\infty}\sum_{n=1}^{\infty} \frac{1}{r}\,\mathrm{Ai}\left(-\left(\lambda_{n\ell}^2/R\right)^{1/3} r\right) P_\ell(\cos\theta)\left[A_{n\ell}\cos(\omega_{n\ell} t) + B_{n\ell}\sin(\omega_{n\ell} t)\right]$$

with $\omega_{n\ell} = \lambda_{n\ell} V$.

Part IV — Casimir Boundary-Value Problem on Cosmological Scales

4.1 The Reduced Equation

The 1+1 reduced (flat-space proxy) form of the gamma field equation:

$$\frac{1}{V^2}\frac{\partial^2 f}{\partial t^2} - \frac{\partial^2 f}{\partial r^2} = 0$$

Boundary conditions: $f(2R,\,0) = 0, \qquad f(2R_s,\,0) = 0$

$$2R = 7.8456 \times 10^{22}\ \text{m}, \qquad 2R_s = 1.5533 \times 10^{26}\ \text{m}$$

4.2 Static Polynomial Solution

A degree-2 polynomial ansatz (separable, static particular solution) satisfying (C):

$$f(r,t) = \frac{(-4R^2 + 2Rr)V^2 + (-4R^2 + r^2)\,c_2}{8\pi r^2 V^2}$$

4.3 Determining c_2 from Boundary Conditions

Condition $f(2R, 0) = 0$ is satisfied identically (numerator vanishes). ✓

Condition $f(2R_s, 0) = 0$: setting the numerator to zero at $r = 2R_s$:

$$4R(R_s - R)V^2 + 4(R_s - R)(R_s + R)\,c_2 = 0$$

Dividing by $4(R_s - R) \neq 0$: $\boxed{c_2 = -\frac{R\,V^2}{R_s+R}}$

4.4 Numerical Evaluation

$$R_s + R = 7.8057 \times 10^{25}\ \text{m}, \qquad V^2 = 8.9875 \times 10^{34}\ \text{m}^2\,\text{s}^{-2}$$

$$c_2 = -\frac{3.9228 \times 10^{22} \times 8.9875 \times 10^{34}}{7.8057 \times 10^{25}}$$

$$\boxed{c_2 \approx -4.5164 \times 10^{31}\ \text{m}^2\,\text{s}^{-2}}$$

4.5 Explicit Static Solution

Cancelling V^2 from numerator and denominator:

$$\boxed{f(r) = \frac{1}{8\pi r^2}\left[-\frac{4R^2R_s}{(R_s+R)} + 2R\,r - \frac{R}{R_s+R}\,r^2\right]}$$

Equivalently, in three separate terms:

$$f(r) = \frac{1}{8\pi}\left[-\frac{4R^2R_s}{(R_s+R)r^2} + \frac{2R}{r} - \frac{R}{R_s+R}\right]$$

4.6 Numerator Polynomial Coefficients $N(r) = \alpha_0 + \alpha_1 r + \alpha_2 r^2$

$$\alpha_0 = -\frac{4R^2R_s}{R_s+R} \approx -6.1242 \times 10^{45}\ \text{m}^2$$

$$\alpha_1 = 2R = 7.8456 \times 10^{22}\ \text{m}$$

$$\alpha_2 = -\frac{R}{R_s+R} \approx -5.0257 \times 10^{-4}\ \text{(dimensionless)}$$

Part V — Particle Flux and Casimir Radiation Pressure 5.1 Gamma-Ray Longitudinal Flux Density

$$\Phi(r) = \frac{f(r)}{4\pi r^2} = \frac{\alpha_0 + \alpha_1 r + \alpha_2 r^2}{32\pi^2 r^4}$$

The $\sim r^{-4}$ falloff at large r is steeper than the photon $\sim r^{-2}$ flux, consistent with the longitudinal (non-transverse) polarisation of the gamma field.

5.2 Quantised Eigenfrequencies

From Part III: $\omega_n = \lambda_n V = \frac{a_n^{3/2} V}{2^{3/2} R}$

5.3 Casimir Radiation Pressure

The longitudinal field stress-energy generates a Casimir-type vacuum pressure:

$$\mathcal{P}_{\text{Casimir}} = \frac{\hbar V}{2L}\sum_n \omega_n\,, \qquad L = 2R_s - 2R = 2(R_s - R)$$

The net inward Casimir force per unit area:

$$F_{\text{Casimir}} = -\frac{\partial \mathcal{P}_{\text{Casimir}}}{\partial L} \propto \frac{\hbar V}{L^2}$$

With $V = 10^9 c$, the Casimir effect is enhanced by a factor of 10^9 relative to the electromagnetic vacuum, reflecting the denser longitudinal mode spectrum at superluminal phase velocity.

Part VI — Comparative Summary 6.1 Gamma vs. Photon: Field Properties

Property	Transverse photon	Longitudinal gamma (TauTimp)
Polarisation	$\perp \boldsymbol{k}$	$\parallel \boldsymbol{k}$
Phase velocity	c	$V = 10^9 c$
Field equation	$\Box A^\mu = 0$	$(R/r)\,\Delta f = (1/V^2)\,\partial_{tt} f$
Mode structure	Plane waves	Airy eigenmodes
Planck constant	$h = 4.136 \times 10^{-15}$ eV·s	$h_\gamma = 9.341 \times 10^{-18}$ eV·s
Flux falloff	r^{-2}	r^{-4}
Casimir scaling	$\hbar c/L^2$	$\hbar V/L^2 = 10^9\,\hbar c/L^2$
Compton shift	$\lambda_C = 2.426$ pm	$\lambda_{C\gamma} = 5.48$ fm

6.2 Key Numerical Results

Quantity	Symbol	Value
Time quantum	T_q	1.765×10^{-19} s
Gamma Planck constant (eq. 6)	h_γ	9.3405×10^{-18} eV·s
Gamma Planck constant (Davisson)	$h_{1\gamma}$	9.570×10^{-18} eV·s
Consistency error		$< 2.4\%$
Gamma wavelength at 54 eV	λ_γ	377 fm
Modified Compton shift	$\Delta\lambda_\gamma$	5.48 fm
Casimir constant	c_2	$-4.5164 \times 10^{31}\ \text{m}^2\,\text{s}^{-2}$

Summary of Key Equations

$$h_\gamma = \frac{32\pi^2 E_0 T_q}{3} \approx 9.341 \times 10^{-18}\ \text{eV·s}$$

$$c_2 = -\frac{R\,V^2}{R_s + R} \approx -4.5164 \times 10^{31}\ \text{m}^2\,\text{s}^{-2}$$

$$f(r) = \frac{1}{8\pi r^2}\left[-\frac{4R^2 R_s}{R_s + R} + 2R\,r - \frac{R}{R_s + R}\,r^2\right]$$

$$\lambda_n = \frac{a_n^{3/2}}{2^{3/2}R}, \qquad \omega_n = \lambda_n V = \frac{a_n^{3/2} V}{2^{3/2}R}$$

$$H(r,\theta,t) = \sum_{n,\ell} \frac{\text{Ai}\left(-(\lambda_{n\ell}^2/R)^{1/3} r\right)}{r}\, P_\ell(\cos\theta)\left[A_{n\ell}\cos\omega_{n\ell}t + B_{n\ell}\sin\omega_{n\ell}t\right]$$

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